

AEGIS: Structural Sensitivity for Adversarial Vulnerability Analysis of GNNs

Anonymous Author(s)

Abstract—Graph neural networks deployed in safety-critical settings (drug interaction, fraud detection, power-grid contingency screening) are vulnerable to adversarial graph perturbations, but no existing method unifies the three diagnostic outputs a practitioner needs from a single closed-form computation: a globally maximally sensitive perturbation direction, per-edge vulnerability rankings, and per-node first-order sensitivity radii. We introduce AEGIS, which derives all three from one object, the *constrained sensitivity matrix* S_c , computed matrix-free via a Neumann-series resolvent and randomized SVD that scale to $N=7,650$ on a single GPU. The analytical construction applies to any GNN with continuous edge-weight-modulated message passing (GCN, SAGE, GIN, APPNP, IGNN, edge-weighted GAT[†]); predictive transfer is architecture-dependent (deeper-than-2-layer models transfer most reliably). For the spectral-norm-constrained IGNN subclass (which costs $\sim 6\%$ test accuracy on Cora relative to unconstrained GNNs) we additionally derive a critical perturbation budget $\varepsilon_{\text{crit}}=(1-\kappa)/\|W\|_2$ and a *worst-case* three-regime characterisation along the leading sensitivity direction. The first-order envelope is tight (1.00 ± 0.01) at $\varepsilon=0.01$ across 7 architectures and 9 datasets and loosens to ≤ 1.39 at $\varepsilon=0.20$, the SVD direction inflicts $2\text{--}8\times$ more damage than random, and continuous-to-discrete transfer is positive in 29 of 33 architecture-dataset cells (full-graph Kendall $\tau \in [-0.04, +0.89]$). As a proof-of-concept cross-domain demonstration on IEEE power-flow benchmarks, S_c scores correlate with N-1 contingency rankings on case14–118 ($\tau=+0.37$ to $+0.62$, P@10 = 0.66–0.81); code will be released under tiered access.

Index Terms—Graph neural networks, adversarial robustness, structural sensitivity, deep equilibrium models, graph perturbation, implicit function theorem

I. INTRODUCTION

Graph neural networks are now deployed in domains where structural errors carry tangible consequences: fraudulent accounts evade detection when an adversary inserts edges into a financial graph [1], drug-interaction models misfire under perturbed molecular graphs [2], and power grids can miss critical contingencies when their graph representations are structurally fragile [3]. A practitioner deploying a GNN faces a question current tools leave unanswered: which edges, if perturbed, would cause predictions to fail, and by how much?

Structural attacks [1], [4], [5] produce per-edge surrogate gradients but no globally optimal continuous direction or per-node budgets; randomized smoothing [6], [7], [8], [9] and IBP-style certifiers [10], [11] give per-node certificates but no edge rankings or direction; robust-architecture defenses [12], [13], [14] harden models without surfacing a vulnerability map, and recent sober-look critiques [15], [16] flag evaluation pitfalls. We unify the three outputs at the level of the analytical object S_c ; predictive ranking transfer is quantified per architecture in table VIII.

We introduce AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity), built around one object: S_c (section IV). For contractive implicit GNNs the construction admits formal regime guarantees (theorem 1); for any GNN with continuous edge-weight-modulated message passing it is a computational tool with empirically validated first-order tightness at small ε (section V-F).

Contributions. (1) S_c and a matrix-free pipeline yielding the SVD-optimal direction, per-edge rankings, and per-node radii from one computation, scaling to $N=7,650$, with matrix-free / dense σ_1 agreement to 0.03% at the shared $N=200$. (2) A worst-case three-regime characterisation along the top singular vector of $\hat{A}$ for the spectral-norm-constrained IGNN subclass with critical budget $\varepsilon_{\text{crit}}=(1-\kappa)/\|W\|_2$ (theorem 1); the formal track applies only to this subclass, which costs $\sim 6\%$ accuracy. (3) Empirical evaluation over 9 datasets, 7 architectures, 4 domains (330 runs), with four-quadrant attack comparison, head-to-head benchmarks vs. GR-BCD / PR-BCD [5] / AGNNCert / LODF (table VI), continuous-to-discrete transfer (fig. 5, table VIII), and a power-grid case study (section VI).

II. PRELIMINARIES AND THREAT MODEL

Let $G = (V, E, A)$ be an undirected graph with $N = |V|$ nodes and adjacency $A \in \mathbb{R}^{N \times N}$, node features $X \in \mathbb{R}^{N \times d_{\text{in}}}$, and symmetric normalized adjacency $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$ (self-loops included). Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i -th singular value, and $\text{vec}(\cdot)$ the column-major vectorization. The sensitivity objects introduced below are: the unconstrained $S \in \mathbb{R}^{Nd \times N^2}$, constrained $S_c \in \mathbb{R}^{Nd \times |E|}$, unrolled $S_K \in \mathbb{R}^{Nd \times N^2}$ (theorem 7), and per-node block S_v ; the contractivity $\kappa = \|J_z\|_2$, spectral radius $\rho(J_z)$, pseudospectral index $\eta \geq 1$, critical budget $\varepsilon_{\text{crit}}$, and per-node radius r_v .

IGNN and IFT. A DEQ-GNN [17], [18] iterates a weight-tied F_θ to a fixed point; the IGNN instantiation [19] uses

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}}), \quad (1)$$

with $W \in \mathbb{R}^{d \times d}$ spectral-normalized [20] so $\kappa \leq \|\hat{A}\|_2 \|W\|_2 < 1$, ReLU ϕ , and a learned projection X_{proj} . Training uses implicit differentiation through the fixed point [21], [22] with optional Jacobian regularization [23], and a linear head $f(z^*) = W_{\text{cls}}z^* + b$ produces class predictions. At equilibrium, $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A , with IFT sensitivity

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p} \Big|_{z^*}, \quad (2)$$

where $\|(I - J_z)^{-1}\|_2 \leq 1/(1-\kappa)$ via the convergent Neumann series, and the pseudospectral index $\eta = \|(I - J_z)^{-1}\|_2 (1 - \rho) \geq 1$ [24] measures the gap to the spectral-radius bound (with $\eta=1$ for normal J_z). The IFT has been used for hyperparameter optimization [25] and implicit differentiation [21], but not for adversarial structural analysis.

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \|\delta\hat{A}\|_F \leq \varepsilon, \quad (3)$$

with $\delta\hat{A}$ symmetric ($\delta\hat{A} = \delta\hat{A}^\top$, preserving the undirected graph), edge-only ($[\delta\hat{A}]_{ij} = 0$ for $(i, j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. Discrete insertions or deletions change the topology and require recomputing $D^{-1/2}$; they lie outside the formal guarantee but are connected to the continuous model through theorem 6. Unlike input-feature perturbation [26], [27], structural perturbation changes the operator itself: z^* depends on A through both $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $J_A = \partial F / \partial \text{vec}(A)$, which is what makes structural sensitivity qualitatively different and motivates the construction of S_c in section III.

III. ADVERSARIAL EQUILIBRIUM THEORY

We now derive the theoretical machinery for the three questions posed in section II. The main result (theorem 1) characterizes how vulnerability evolves with perturbation magnitude, and the constrained sensitivity matrix S_c provides the unified object from which rankings, attacks, and radii follow.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs). *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

- (A1) ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT [28] (proof below).
- (A2) $\|W\|_2 \leq c$ via spectral normalization.
- (A3) $\|J_z\|_2 \leq \kappa < 1$ verified post-training ($\kappa=0.14$ – 0.59 across datasets; section V-E). (A2) does not imply (A3) automatically; we report κ alongside $\varepsilon_{\text{crit}}$ in our experiments so practitioners can audit. If (A3) fails, AEGIS still returns S_c rankings and SVD-optimal attacks but flags that the formal guarantees below are void.

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1 - \kappa}{\|W\|_2}. \quad (4)$$

Then structural perturbations $\delta\hat{A}$ with $\|\delta\hat{A}\|_F = \varepsilon$ exhibit the following three regimes, where the critical-regime statement (b) is a worst-case bound along directions aligned with the top singular vector of $\hat{A}$; generic directions stay subcritical longer:

(a) **Subcritical** ($\varepsilon < \varepsilon_{\text{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \quad (5)$$

where $S = (I - J_z)^{-1} J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\text{op}} / (1 - \kappa)$.

(b) **Critical** ($\varepsilon \rightarrow \varepsilon_{\text{crit}}$): along worst-case directions (aligned with the top singular vector of $\hat{A}$), $\|(I - J'_z)^{-1}\|_2$ diverges as $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$; generic directions diverge more slowly since $\|J'_z\|_2$ need not approach 1.

(c) **Supercritical** ($\varepsilon > \varepsilon_{\text{crit}}$): the contraction certificate is void, Banach no longer guarantees uniqueness, and the part-(a) first-order guarantees lapse.

Proof. (a). Under (A1), F_θ is piecewise-affine on each ReLU linear region, so the conservative-gradient calculus of Bolte–Pauwels [28] gives a conservative IFT on each region with a measure-zero exceptional set of activation-boundary crossings. With $J_z = \text{diag}(\phi') \cdot (\hat{A} \otimes W)$ and $\kappa \leq \|\hat{A}\|_2 \|W\|_2$, applying the IFT to $G(z, \hat{A}) = z - F(z, \hat{A}) = 0$ at $(z^*, \hat{A})$ gives

$$\Delta z^* = (I - J_z)^{-1} J_A \cdot \text{vec}(\delta\hat{A}) + O(\|\delta\hat{A}\|_2^2), \quad (6)$$

with $J_A = \partial F / \partial \text{vec}(A)$ and the Neumann bound $\|(I - J_z)^{-1}\|_2 \leq 1/(1 - \kappa)$ yielding (5). For contractivity preservation, $\|J'_z\|_2 \leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2$, so $\varepsilon < \varepsilon_{\text{crit}}$ guarantees $\|J'_z\|_2 < 1$. The Frobenius-operator gap leaves $\varepsilon_{\text{crit}}$ conservative by up to $\sqrt{\text{rank}(\delta\hat{A})}$; the constrained projection S_c tightens this in practice.

(b)–(c). $\|J'_z\|_2 \rightarrow 1$ forces $\|(I - J'_z)^{-1}\|_2 \geq 1/(1 - \|J'_z\|_2) \rightarrow \infty$ regardless of J_z 's normality; the operator-norm bound holds whenever $\kappa < 1$. Empirically $\eta=1.02$ – 1.28 (section V-E). When $\|J'_z\|_2 \geq 1$ the contraction-mapping hypothesis fails and the iteration can converge to a different equilibrium, oscillate, or diverge. $\square$

$\varepsilon_{\text{crit}}$ is a sufficient, not necessary, condition: the perturbed system may stay contractive beyond $\varepsilon_{\text{crit}}$ along low-sensitivity directions, and on 50-node subgraphs the unconstrained $\sqrt{r}$ slack is 7–14 $\times$; S_c closes this gap in practice (tightness ≈ 1.00 at $\varepsilon=0.01$; section V-A). The Neumann series requires the operator-norm κ rather than $\rho(J_z)$ [29]; their gap is the pseudospectral index η , which is bounded by the weight matrix alone:

Observation 2 (Graph-Independent Nonnormality Bound). *For $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ with symmetric $\hat{A}$: if all activations are positive, $\eta \leq \kappa(V_W)$ where $W = V_W D_W V_W^{-1}$ and $\kappa(V_W) = \|V_W\|_2 \|V_W^{-1}\|_2$, independent of $\hat{A}$. For general ReLU patterns $\eta \leq \kappa(V_{J_z})$, which empirically stays near $\kappa(V_W)$ ($\eta=1.02$ – 1.28 ; section V-E).*

Proof. When $\phi' \equiv 1$, $J_z = \hat{A} \otimes W$ has Kronecker eigenvalues $\lambda_i(\hat{A})\lambda_j(W)$ [29]; symmetric $\hat{A}$ gives $\kappa(U_{\hat{A}})=1$, so the joint eigenvector matrix has condition number $\kappa(V_W)$, and the diagonalisable bound $\|(I - M)^{-1}\|_2 \leq \kappa(V)/(1 - \rho(M))$ yields $\eta \leq \kappa(V_W)$. Graph structure does not amplify nonnormality; the mild empirical η traces to the spectral-norm regularisation of W .

Proposition 3 (Maximally Sensitive First-Order Structural Perturbation). *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first*

order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot \text{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S . The maximum first-order equilibrium shift is $\varepsilon \cdot \sigma_1(S)$.

The result follows from the variational characterisation of the largest singular value; the direction is optimal for hidden-state damage and empirically aligned with classification damage (section V-B). The constrained sensitivity matrix $S_c \in \mathbb{R}^{Nd \times |E|}$ has columns $[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i}$ paired with the edge basis $b_k = (e_i e_j^\top + e_j e_i^\top) / \sqrt{2}$ (so $\|b_k\|_F = 1$ and $\sigma_1(S_c)$ is consistent with $\|\delta \hat{A}\|_F = \varepsilon$). The $N^2 \rightarrow |E|$ projection P_c is the standard duplication-matrix reduction [30] restricted to the edge-supported symmetric subspace; the operational contribution is the matrix-free $S_c v$ in algorithm 1. Empirically $\sigma_1(S_c)$ gives a first-order envelope ratio ≈ 1.00 at $\varepsilon = 0.01$; the ratio rises to ≤ 1.39 at $\varepsilon = 0.20$, so the bound is tight only at small ε and identifies the dangerous direction rather than a global safety margin.

Proposition 4 (Per-Node First-Order Sensitivity Radius). *For a linear head $f(z) = Wz + b$ and node v with margin $m_v = f_{y_v}(z_v^*) - \max_{c \neq y_v} f_c(z_v^*) > 0$, the first-order sensitivity radius*

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2} \quad (7)$$

preserves the classification of v for any $\|\delta \hat{A}\|_F < r_v$, where S_v are the block-rows of S at node v . Under the constrained threat model the tighter radius $r_v^{(c)} = m_v / (\|W_{y_v} - W_{c^}\|_2 \|S_{c,v}\|_2) \geq r_v$ holds, using the block-rows of S_c at node v .*

Proof sketch. By Thm. 1(a), $\Delta z_v^* \approx S_v \cdot \text{vec}(\delta A)$, so $|\Delta f_{y_v}| \leq \|W_{y_v} - W_{c^*}\|_2 \|S_v\|_2 \|\delta A\|_F$; misclassification requires $|\Delta f_{y_v}| \geq m_v$, and inverting gives r_v . Intuitively, r_v is the margin divided by two amplification factors (decision-boundary sensitivity to hidden-state change, and equilibrium sensitivity to structural perturbation); high-margin, low-sensitivity nodes get large radii.

Remark 5 (Sensitivity threshold semantics). r_v is a first-order sensitivity threshold, not a probabilistic certificate; it indicates the perturbation scale at which the linearization predicts a classification change for node v , and is locally tight for small ε but can be violated at larger magnitudes where higher-order terms dominate. The two approaches are complementary: randomized smoothing [6] certifies which nodes are guaranteed safe, while AEGIS differentiates which edges and nodes are most sensitive and in which direction (section V-B). Empirically, breach rate is 0% at $\varepsilon = 0.01$ and rises with ε as expected for first-order approximations (section V-B).

A. Continuous-to-Discrete Transfer

Proposition 6 (Continuous-to-Discrete Ranking Transfer). *Under (A1)–(A3), let $w_k = [\hat{A}]_{ij}$ denote the normalized weight of edge $k = (i, j) \in E$, $v_k = \| [S_c]_{:,k} \|_2$ the continuous vulnerability score, and $d_k = \|z^*(A) - z^*(A \setminus k)\|_2$ the*

discrete damage from full removal of edge k . Assume all single-edge removals are subcritical: $\sqrt{2} \max_k w_k < \varepsilon_{\text{crit}}$.

(a) First-order bridge. *The discrete damage satisfies*

$$d_k = w_k \cdot v_k + R_k, \quad |R_k| \leq \frac{L_J}{2(1-\kappa)^2} \cdot w_k^2, \quad (8)$$

where L_J is the Lipschitz constant of the state Jacobian J_z with respect to $\text{vec}(A)$.

(b) Ranking preservation. *For two edges k_1, k_2 with $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$, we have $d_{k_1} > d_{k_2}$ whenever*

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J / (1-\kappa)^2}. \quad (9)$$

Under uniform weights ($w_k = w$ for all k), $v_{k_1} > v_{k_2}$ implies $d_{k_1} > d_{k_2}$ whenever $w < \min_{k_1 \neq k_2} (v_{k_1} - v_{k_2}) \cdot (1-\kappa)^2 / L_J$.

Proof. **(a).** Edge removal corresponds to $[\delta \hat{A}]_{ij} = [\delta \hat{A}]_{ji} = -w_k$, so by the S_c construction $\|\Delta z^*\| \approx w_k v_k$. For the remainder, let $g(t) = z^*(A + t\delta \hat{A}_k)$ with $g'(0) = S \cdot \text{vec}(\delta \hat{A}_k)$. Taylor expansion, the resolvent identity $\frac{d}{dt}(I - M)^{-1} = (I - M)^{-1} \frac{dM}{dt} (I - M)^{-1}$, and Lipschitz continuity $\|dJ_z/dt\|_2 \leq L_J \|\delta \hat{A}_k\|_2$ (with $L_J \leq \|W\|_2^2$ for IGNN) yield $\|g''(\xi)\| \leq (1-\kappa)^{-2} L_J w_k^2$, so $|R_k| \leq L_J w_k^2 / (2(1-\kappa)^2)$. **(b).** From (a), $d_{k_1} - d_{k_2} \geq w_{k_1} v_{k_1} - w_{k_2} v_{k_2} - C(w_{k_1}^2 + w_{k_2}^2)$ with $C = L_J / (2(1-\kappa)^2)$; under $v_{k_1} > v_{k_2}$, $w_{k_1} \geq w_{k_2}$, the first term is positive and (9) ensures the first-order gap dominates. $\square$

For ReLU activations, J_z is locally constant in each linear region (Taylor remainder exact, $R_k = 0$) and jumps by at most $\|W\|_2^2$ across activation boundaries, so (8) holds with $L_J \leq \|W\|_2^2$. Empirically $|R_k|$ is 2–10 $\times$ smaller than $L_J w_k^2$, and the 29/33 positive-transfer cells of fig. 5 confirm the piecewise structure does not erode the bridge: shallow models with near-uniform v_k (GCN-2) violate part-(b), deeper models satisfy it. The ingredients of theorem 1 are classical [29], [24]; the contribution is the constrained projection S_c that turns a vacuous bound (tightness 0.31) into a tight one (1.00; section V-A) and unifies attacks, rankings, and radii.

B. Generalization to Explicit GNNs

Observation 7 (Structural Sensitivity for K -Layer GNNs). *Let $Z_K = g_K \circ \dots \circ g_1(X, A)$ be a K -layer GNN where each layer g_l is differentiable with respect to both its input and A . Define the unrolled sensitivity matrix*

$$S_K = \frac{\partial \text{vec}(Z_K)}{\partial \text{vec}(A)} = \sum_{l=1}^K \left(\prod_{k=l+1}^K J_z^{(k)} \right) J_A^{(l)}, \quad (10)$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \text{vec}(A)$. Then:

(a) *The first-order shift bound $\|\Delta Z_K\|_F \leq \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with*

$$\sigma_1(S_K) \leq \sum_{l=1}^K \left(\prod_{k=l+1}^K \|J_z^{(k)}\|_2 \right) \|J_A^{(l)}\|_2. \quad (11)$$

(b) *The constrained construction S_c and per-node radius r_v (theorem 4) apply identically with S_K in place of S .*

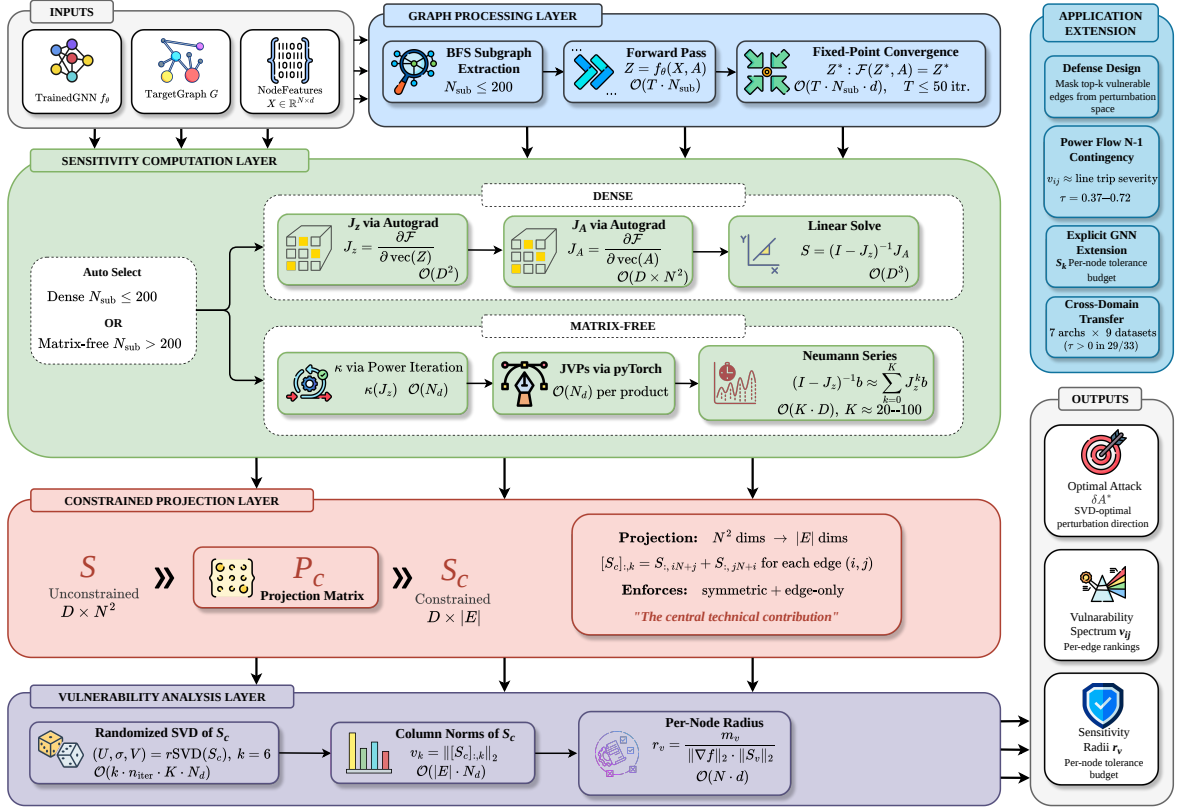


Fig. 1: The four-stage AEGIS pipeline: (1) graph processing (BFS subgraph or full-graph forward / fixed-point pass); (2) sensitivity computation (dense Jacobian for $N \leq 200$, matrix-free Neumann/JVP otherwise); (3) constrained projection $N^2 \rightarrow |E|$ via P_c ; (4) vulnerability analysis (randomised SVD, S_c column norms, per-node radii). The matrix-free path queries S_c via JVPs/VJP without materialising it, scaling to $N=7,650$.

For weight-tied models, $\sigma_1(S_K) \leq \|J_A\|_2 (1 - \kappa^K) / (1 - \kappa)$ converges to the IGNN bound as $K \rightarrow \infty$. Explicit GNNs inherit the practical tools but lack $\varepsilon_{\text{crit}}$ and the regime characterisation, which depend on (A3); a Jacobian product $\prod_k \|J_z^{(k)}\|_2 > 10$ diagnoses unreliable first-order predictions, and the empirical tightness 0.99–1.02 (section V-F) validates the approximation regardless of the a priori bound.

IV. AEGIS CONSTRUCTION

The pipeline turns section III into a practical tool through matrix-free computation, sidestepping both the materialisation of $S \in \mathbb{R}^{Nd \times N^2}$ and the formation of the resolvent. One S_c computation returns the per-edge spectrum $\{v_{ij}\}$, the SVD-optimal direction $\delta \hat{A}^*$, and per-node radii $\{r_v\}$, plus $\varepsilon_{\text{crit}}$ and the regime characterisation of theorem 1 for IGNN-class operators. A dense path ($N \leq 200$) computes an exact SVD; a matrix-free path ($N > 200$, validated to $N=7,650$) uses JVP/VJP queries and randomized SVD. Figure 1 illustrates the four stages; algorithm 1 summarises the matrix-free routing.

Matrix-free operator. $S_c v = (I - J_z)^{-1} J_A P_c v$ is evaluated via the truncated Neumann series $\sum_{k=0}^K J_z^k b$ with depth K adaptive in κ (early stop at $\|J_z^k b\| < 10^{-6} \|b\|$); each Neumann term and $J_A \cdot \text{vec}(\delta A)$ are forward-mode JVPs in $\mathcal{O}(Nd)$ time,

and the adjoint $S_c^\top u$ uses the analogous VJPs. The constrained projection P_c is applied implicitly inside the operator so neither S_c nor S is formed; the dense path ($N \leq 200$) materialises J_z, J_A and solves via `torch.linalg.solve`. The randomized SVD [31] ($k=10, p=10, n_{\text{iter}}=2$) returns (u_1, σ_1, v_1) for $\delta \hat{A}^* = \varepsilon \cdot \text{sym}(P_c v_1)$; column norms give $\{v_{ij}\}$, and per-node radii follow theorem 4 via the node-restricted RMATVEC. The well-separated singular spectrum visible in fig. 2 (gap $(\sigma_1 - \sigma_2) / \sigma_1 = 0.39-0.50$) is why one closed-form SVD matches iterative attackers (section V-B).

Cost. Each $S_c v$ is $\mathcal{O}(K \cdot Nd)$ time and $\mathcal{O}(Nd)$ memory ($K \in [20, 50]$ for $\kappa < 0.8$); removing the $\mathcal{O}((Nd)^3)$ solve lifts the previous $N \approx 300$ ceiling (section V-D).

V. EXPERIMENTS

Setup. We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each, using PyTorch [32]. IGNN [19] uses $F(Z) = \text{ReLU}(\hat{A} Z W^\top + U(X))$ with $W \in \mathbb{R}^{64 \times 64}$ spectral-normalized [20] and Adam (lr 0.01); ContractiveGCN-PF mirrors this with 5 bus features and a 2-head readout. Subgraphs are 50-node BFS from the highest-degree node. The spectral-norm constraint costs IGNN $\sim 6\%$ accuracy (77.5% vs. $\sim 83\%$ on Cora), the price of the guarantees in

Algorithm 1 AEGIS Vulnerability Analysis (matrix-free). Assumes the equilibrium z^* is pre-computed on the analysis (sub)graph. Dense fallback ($N \leq 200$): form $S = (I - J_z)^{-1} J_A$ and run a deterministic SVD in place of lines 2–11.

Require: $F, z^*, \hat{A}$, budget ε ; SVD rank k , oversampling p , power iters n_{iter} ; Neumann tol. τ ; edge set E ; (opt.) head h , logits Y , labels y .

Ensure: $\sigma_1(S_c), \delta \hat{A}^*, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}}$.

- 1: $\kappa \leftarrow \|J_z\|_2$ via power iteration; $K \leftarrow \lceil \log(1/\tau) / \log(1/\kappa) \rceil$.
- 2: $\text{MATVEC}(v) := \sum_{j=0}^K J_z^j \text{JVP}_{\hat{F}}^{\hat{A}}(P_c v)|_{z^*}$;
 $\text{RMATVEC}(u) := P_c^\top \text{VJP}_{\hat{F}}^{\hat{A}}(\sum_{j=0}^K (J_z^\top)^j u)|_{z^*}$.
- 3: Draw $\Omega \sim \mathcal{N}(0, I) \in \mathbb{R}^{|E| \times (k+p)}$; $Y \leftarrow \text{MATVEC}(\Omega)$ column-wise.
- 4: **for** $j = 1, \dots, n_{\text{iter}}$ **do**
- 5: $Y \leftarrow \text{QR}(\text{MATVEC}(\text{RMATVEC}(Y)))$
- 6: **end for**
- 7: $Q \leftarrow \text{QR}(Y)$; $B^\top \leftarrow \text{RMATVEC}(Q)$; $(\tilde{U}, \Sigma, V^\top) \leftarrow \text{SVD}(B)$; $U \leftarrow Q \tilde{U}$ {rSVD [31]}
- 8: $\sigma_1 \leftarrow \Sigma_{1,1}$; $\delta \hat{A}^* \leftarrow \varepsilon \cdot \text{sym}(P_c V_{:,1}) / \|\text{sym}(P_c V_{:,1})\|_2$.
- 9: $v_{ij} \leftarrow \|\text{MATVEC}(e_{ij})\|_2$ for each $(i, j) \in E$ {per-edge spectrum}
- 10: **if** h, Y, y supplied **then**
- 11: $r_v \leftarrow m_v / (\|W_{y_v} - W_{c^*}\|_2 \|S_{c,v}\|_2)$ for each v via 10 power iters {Prop. 4}
- 12: **end if**
- 13: **if** $\|W\|_2$ extractable **then** $\varepsilon_{\text{crit}} \leftarrow (1 - \kappa) / \|W\|_2$ {Thm. 1}
- 14: **return** $(\sigma_1, \delta \hat{A}^*, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}})$.

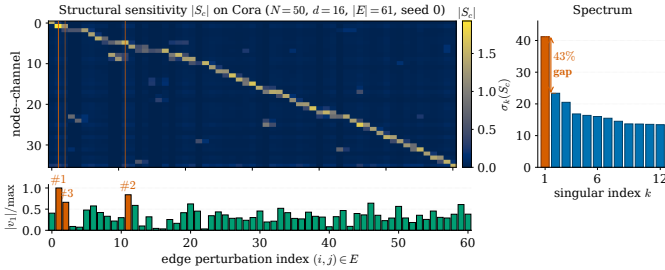


Fig. 2: Structural sensitivity matrix $|S_c|$ on a 50-node Cora ego-graph (IGNN, seed 0). Rows: node-channels; columns: edge perturbations. Vermilion lines mark the top-3 edges by $|v_1|$ (SVD-optimal attack direction, bar strip below); sidebar: singular spectrum with $\sigma_1 = 41.2$ and a 43% gap to σ_2 , matching the cross-benchmark $(\sigma_1 - \sigma_2) / \sigma_1 \in [0.39, 0.50]$ band of section V-B.

theorem 1. Datasets: Cora, Citeseer, Pubmed (Planetoid [33]), Amazon Photo [34], WikiCS [35], and IEEE test cases 14/30/57/118 [36], [37]. Baselines: smoothing (Gaussian $\sigma = 0.01$, 200 MC, Clopper-Pearson $\alpha = 0.001$), Mettack with Meta-Self GCN surrogate, and LODF as $\text{PTDF}_{ij} / (1 - \text{PTDF}_{jj})$ from the DC B -matrix [38, Ch. 11]. Code (dense +

TABLE I: Per-dataset IGNN characterisation (10 seeds). $\kappa = \|J_z\|_2$ verifies (A3); $\varepsilon_{\text{crit}} = (1 - \kappa) / \|W\|_2$ is the theoretical critical budget. AtkAdv: AEGIS damage / random damage. Cov%: fraction of subgraph nodes with positive first-order radius r_v .

Dataset	Test Acc	κ	$\varepsilon_{\text{crit}}$	AtkAdv	Cov%
Cora	77.5 $\pm$ 1.7	.33 $\pm$.14	.66 $\pm$.15	3.6 $\pm$.6	81 $\pm$ 9
Citeseer	66.0 $\pm$ 0.7	.59 $\pm$.02	.41 $\pm$.02	4.1 $\pm$.5	83 $\pm$ 4
Pubmed	78.9 $\pm$ 0.7	.41 $\pm$.11	.59 $\pm$.11	3.2 $\pm$.4	74 $\pm$ 23
Amazon	94.8 $\pm$ 0.3	.14 $\pm$.02	.86 $\pm$.02	3.8 $\pm$.2	92 $\pm$ 4
WikiCS	77.9 $\pm$ 0.4	.34 $\pm$.04	.66 $\pm$.04	3.8 $\pm$.4	70 $\pm$ 3

TABLE II: First-order envelope ratio (actual / first-order-predicted equilibrium shift) with increasing ε (10 seeds, S_c -optimal constrained perturbation). Tight at $\varepsilon = 0.01$ (≈ 1.00); under-predicts by up to 39% at $\varepsilon = 0.20$.

Dataset	$\varepsilon = .01$	$\varepsilon = .05$	$\varepsilon = .10$	$\varepsilon = .20$
Cora	1.01 $\pm$.00	1.07 $\pm$.01	1.15 $\pm$.03	1.36 $\pm$.08
Citeseer	1.01 $\pm$.00	1.07 $\pm$.01	1.16 $\pm$.03	1.39 $\pm$.07
Pubmed	1.01 $\pm$.00	1.07 $\pm$.01	1.15 $\pm$.02	1.38 $\pm$.06
WikiCS	1.00 $\pm$.01	1.01 $\pm$.01	1.02 $\pm$.01	1.05 $\pm$.01
Amazon	1.00 $\pm$.00	1.01 $\pm$.00	1.03 $\pm$.01	1.07 $\pm$.01

matrix-free S_c , training scripts, exact 10 seeds, dataset splits, PandaPower contingency script, version pins) and trained model checkpoints will be released under tiered access with attack-generation behind institutional-affiliation review.

A. Cross-Domain Adversarial Analysis

Unless otherwise stated, IGNN experiments in section V-A–section V-E use 50-node BFS ego-subgraphs centred on the highest-degree node and report *local vulnerability*; on Cora ($N = 2,708$), 50-node rankings correlate weakly with full-graph rankings ($\tau = 0.16$; section V-E), so graph-level assessment uses the matrix-free pipeline (section V-D). All bounds use the operator-norm $\kappa = \|J_z\|_2$; $\eta = 1.02$ – 1.28 keeps the κ -based bound at most 28% more conservative than the spectral-radius bound. Table I verifies (A3) on all benchmarks ($\kappa < 1$) and reports the per-dataset critical budget $\varepsilon_{\text{crit}}$ and attack advantage.

Table II reports tightness degradation across the five datasets and four ε values.

The envelope ratio is tight (1.00–1.02) at $\varepsilon = 0.01$ and grows to 1.05–1.39 at $\varepsilon = 0.20$; the bound is reliable only at small ε and under-predicts by up to 39% at the largest budget. The ratio is measured on equilibrium shift $\|\Delta z^*\|_F$, distinct from the prediction-flip (breach) rate of table V (flips stay $< 2\%$ at $\varepsilon \leq 0.10$, $< 6\%$ at $\varepsilon = 0.20$).

Heuristic and surrogate-attack baselines. AEGIS beats degree-proportional, edge-betweenness, and spectral rankings at $\varepsilon = 0.01$ (+6–148% AtkAdv, Wilcoxon $p < 0.001$, all 10 seeds) and inflicts 3–10 $\times$ more damage than Mettack [4] (149/150 wins, $k \in \{1, \dots, 5\}$, 3 datasets); GR-BCD / PR-BCD [5] comparisons are in table VI. The continuous gap over degree is small because hubs concentrate message passing;

TABLE III: Equilibrium damage at $\varepsilon=0.10$ across attack methods (IGNN, 50-node subgraph, 10 seeds). SVD: AEGIS. Cls-PGD: classification-loss PGD. Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

Dataset	SVD	Cls-PGD	Shift-PGD	Random
Cora	$3.70 \pm .79$	$2.51 \pm .48$	$3.05 \pm .70$	$0.97 \pm .25$
Citeseer	$4.63 \pm .71$	$2.97 \pm .42$	$3.35 \pm .57$	$1.01 \pm .15$
WikiCS	$2.10 \pm .22$	$0.67 \pm .11$	$1.93 \pm .20$	$0.53 \pm .09$

AEGIS’s distinctive value appears under discrete removal and in the SVD direction itself.

B. Comprehensive Attack Evaluation

Radius vs. smoothing. Against randomized smoothing [6] ($\sigma \in \{0.01, 0.05, 0.10, 0.15\}$, 1000 MC samples, 10 seeds), base smoothing radii are constant per node ($r = \sigma \cdot \Phi^{-1}(p_{\text{lower}})$), e.g. 0.123 vs. AEGIS’s 0.046 at $\sigma=0.05$) and offer no per-edge differentiation; localized smoothing [8] gives per-node deterministic-radius probabilistic certificates at 10^3 – 10^4 MC samples per node but still lacks an edge structure or attack direction. AEGIS radii are structurally informative ($r_v \approx 0.09$ in dense regions, 0.01 at boundaries), placing the three methods at complementary points on the certificate-vs-diagnostic frontier.

Table III evaluates AEGIS against baselines spanning the four quadrants of (gradient-based, gradient-free) $\times$ (equilibrium-shift, classification-loss). Three observations follow: 1,000 random unit directions reach only 0.45 – $0.49 \times$ of $\sigma_1(S_c)$, confirming the SVD direction is well separated; an equilibrium-shift PGD attacker [39] with AEGIS’s IFT gradients (50 steps) recovers only 72–92% of the closed-form damage; and a classification-loss PGD through the unrolled iteration inflicts 15–70% less equilibrium damage while flipping predictions comparably (0–1.8%), so the SVD direction flips predictions effectively even when optimising a proxy. We therefore describe the SVD direction as the *maximally sensitive perturbation direction* rather than “the optimal attack”.

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20 \pm 0.05$ vs. AEGIS’s $+0.32 \pm 0.04$) and stay near zero elsewhere; AEGIS achieves 2–6 $\times$ higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces.

Discrete edge removal. Table IV reports cumulative damage under four sequential orderings. AEGIS matches the greedy proxy on Citeseer and WikiCS and reaches 54–67% on Cora, where cascade effects beat a static ranking; degree-ranked removal is below random on Cora at every k because hub nodes have redundant connections.

Per-edge finite differences reproduce the S_c column-norm ranking ($\tau=0.999$ across datasets), confirming numerical correctness.

Breach rates. Table V reports prediction-flip fractions across ε . Pubmed is right-skewed (median 7.8% vs. mean 10.3% at $\varepsilon=0.10$); every breached node satisfies $\varepsilon > r_v$, consistent

TABLE IV: Cumulative damage from sequential discrete edge removal (IGNN, 50-node subgraph, 10 seeds). Greedy: sequential most-damaging-edge proxy.

Dataset	k	Greedy	AEGIS	Degree	Random
Cora	5	$3.36 \pm .68$	$1.80 \pm .44$	$0.90 \pm .19$	$1.27 \pm .25$
Cora	10	$4.08 \pm .82$	$2.75 \pm .66$	$1.48 \pm .28$	$2.02 \pm .36$
Citeseer	10	$6.01 \pm .94$	$6.01 \pm .94$	$1.93 \pm .26$	$2.46 \pm .56$
WikiCS	10	$0.23 \pm .02$	$0.21 \pm .03$	$0.12 \pm .01$	$0.12 \pm .01$

TABLE V: Breach rate (%) with increasing ε (10 seeds, S_c -optimal constrained perturbation; mean $\pm$ std, Pubmed right-skewed at $\varepsilon \geq 0.05$).

Dataset	$\varepsilon=.01$	$\varepsilon=.05$	$\varepsilon=.10$	$\varepsilon=.20$
Cora	0.6 ± 1.2	0.8 ± 1.3	2.4 ± 3.7	7.6 ± 7.5
Citeseer	0.0 ± 0.0	0.0 ± 0.0	0.2 ± 0.8	1.5 ± 2.4
Pubmed	0.0 ± 0.0	7.0 ± 8.3	10.3 ± 11.0	27.4 ± 21.1
WikiCS	0.3 ± 0.9	0.6 ± 1.3	0.9 ± 1.4	1.8 ± 1.5

with r_v as a first-order diagnostic rather than a probabilistic certificate (theorem 5).

C. Comparison with Prior Structural Baselines

Table VI reports head-to-head numbers against external methods spanning the attacker, certifier, and operational-screener axes: GR-BCD [5] (the same paper introduces PR-BCD for OGB-scale graphs, discussed below), AGNNCert [10], and LODF [38]. AEGIS is a label-free closed-form ranking proxy throughout; the question is how much of each gold-standard signal the proxy retains.

GR-BCD [5] converges with S_c on Pubmed ($\tau=+0.69$) and diverges on Cora ($\tau=+0.16$), the dataset-level τ acting as a self-diagnostic; the labelled attacker dominates AEGIS on Cora absolute damage by $1.9 \times$ at $k=5$, as expected for a label-aware attacker vs. a label-free closed-form proxy. PR-BCD [5] (same family, larger OGB-scale budget) is a natural follow-up benchmark beyond our matrix-free boundary ($N \approx 7,650$). AGNNCert [10]’s IBP radii are 4.9 – $10.2 \times$ looser numerically but mathematically stronger (sound certificate vs. first-order threshold). LODF [38] retargeted to thermal-overload peaks at P@10=0.60 on case57 and collapses on case118; the Ejebe–Wollenberg PI [40] is positively correlated but loses to AEGIS on both grids.

D. Phase Transition and Scalability

Phase transition. Figure 3 sweeps $\kappa_{\text{max}} \in [0.30, 0.99]$ on Cora (10 seeds each): the theoretical $\varepsilon_{\text{crit}}$ drops $230 \times$ but $\rho(J_z)$ saturates at ≈ 0.42 (the ReLU pattern blocks the Jacobian from its theoretical maximum), so the resolvent rises only $1.17 \rightarrow 1.80$ instead of the $\sim 70 \times$ that $1/(1 - 0.99)$ predicts. The phase transition is therefore a theoretical safety boundary, not an empirically observable discontinuity; training converges throughout the sweep, though the post-hoc AEGIS analysis exceeds the 50-step fixed-point budget at $\kappa_{\text{max}} \geq 0.85$.

TABLE VI: AEGIS vs. external baselines (10 seeds). GR-BCD [5] is the BCD-family scalable structural attacker (PR-BCD is the same paper’s larger-budget variant, see text); AGNNCert [10] is a sound IBP certifier (semantics differ from AEGIS r_v); LODF [38] is the industry DC-PF screener.

Baseline	Setting	AEGIS	Baseline	τ
GR-BCD [5]	Pubmed $k=10$	0.356	0.350	+0.69
GR-BCD [5]	Cora $k=5$	0.643	1.207	+0.16
AGNNCert [†] [10]	Cora med. r_v	0.187	1.414	+0.08
LODF [38]	case118 P@10	0.81	≤ 0.20	+0.14

[†]AGNNCert is a *sound* IBP certificate; AEGIS r_v is a first-order sensitivity threshold (theorem 5). The numerical gap reflects different mathematical objects, not framework dominance.

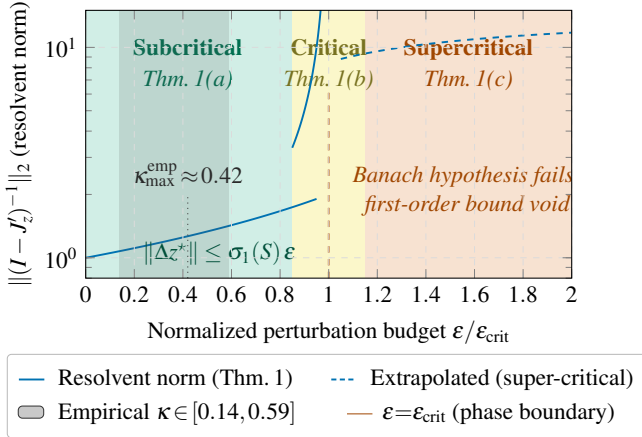


Fig. 3: Three-regime characterisation of theorem 1. The empirical κ band (grey) saturates well below the spectral-normalisation ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1 - \kappa)$ divergence at $\varepsilon \rightarrow \varepsilon_{\text{crit}}$.

Scalability and matrix-free self-consistency. Figure 4 compares the dense and matrix-free pipelines. The matrix-free $S_c v$ operator implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at the shared $N=200$ and per-edge ranking $\tau=0.999$, so the matrix-free rankings are the same algebraic object at scale (not an independent measurement). For larger N the rSVD error [31] is bounded by the spectral gap. Pubmed ($N=19,717$) exceeds 24 GB.

E. Ablations and Defense

Subgraph size. Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$, degrades to 1.031 at $N=200$, coverage stays 82–89%. We use $N=50$ as default ($66\times$ faster than $N=200$, tightness 1.013 ± 0.003). Subgraph-vs-full-graph agreement is high on dense grids (case14 $\tau=0.80$ – 0.86 at $N_{\text{sub}}=8$ – 12 ; case30 $\tau=0.78 \pm 0.07$) but low on citation-scale graphs (Cora $\tau=0.16 \pm 0.13$ since a 50-node BFS covers $\sim 1.8\%$ of edges), so citation-scale users should prefer the full-graph pipeline. BFS-centre choice barely matters ($\tau=0.95 \pm 0.02$ across highest-vs-median-degree).

Hyperparameters. Tightness is stable across hidden width $d \in \{16, 32, 64, 128\}$; the spectral-norm ceiling $c \in \{0.5, 0.9\}$

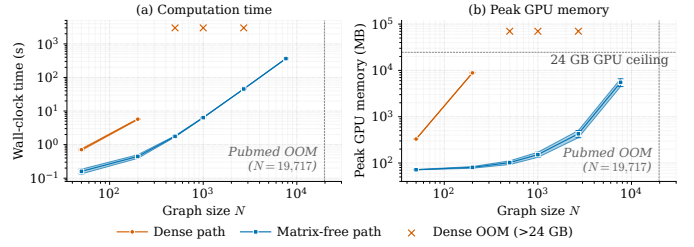


Fig. 4: Dense vs. matrix-free AEGIS pipeline on a single RTX 4090. The dense path scales as $O((Nd)^3)$ and exceeds 24 GB beyond $N=200$; the matrix-free path tracks roughly $O(N \log N)$ in time and grows sub-linearly in memory, reaching Amazon Photo ($N=7,650$) at 365 s / 5.5 GB.

TABLE VII: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ : Kendall vs. brute-force single-edge-removal truth. Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	1.01 $\pm$.00	7.6 $\pm$ 0.5 $\times$	+0.32 $\pm$.04	77.5 $\pm$ 1.7%
GCN-2	1.00 $\pm$.00	3.6 $\pm$ 0.6 $\times$	−.04 $\pm$.03	78.9 $\pm$ 0.9%
GCN-4	1.02 $\pm$.00	3.4 $\pm$ 0.4 $\times$	+0.49 $\pm$.02	78.3 $\pm$ 1.8%
GIN-2	1.01 $\pm$.00	2.6 $\pm$ 0.1 $\times$	+0.33 $\pm$.03	76.6 $\pm$ 1.9%
GAT [†]	1.01 $\pm$.01	2.1 $\pm$ 0.1 $\times$	+0.54 $\pm$.06	77.8 $\pm$ 1.2%
SAGE-2	1.00 $\pm$.00	1.9 $\pm$ 0.2 $\times$	+0.22 $\pm$.10	77.5 $\pm$ 1.5%
APPNP	1.02 $\pm$.00	3.4 $\pm$ 0.3 $\times$	+0.35 $\pm$.01	82.2 $\pm$ 0.6%

traces the accuracy-robustness frontier ($r_v=0.090$ at 72.1% acc vs. $r_v=0.051$ at 80.6%). All runs converge.

Defense-informed edge protection. Masking the top- k edges by v_{ij} before re-running the SVD attack on the reduced S_c (Cora IGNN, $N=50$, 10 seeds) reduces SVD damage by $42 \pm 8\%$ at $k=5$ vs. $11 \pm 6\%$ for random masking, and by $61 \pm 7\%$ vs. $18 \pm 9\%$ at $k=10$ (paired Wilcoxon $p < 0.002$, all 10 seeds favour vulnerability-guided masking), so v_{ij} identifies edges whose protection disproportionately reduces attack surface.

F. Extension to Explicit GNNs

For K -layer explicit GNNs we define the unrolled sensitivity matrix $S_K = \partial \text{vec}(Z_K) / \partial \text{vec}(A)$ via finite differences and construct S_c from S_K identically to the IGNN case (theorem 7). Table VII reports the resulting per-architecture characterisation on Cora; the accuracy-vs-guarantee frontier is explicit: IGNN carries the formal apparatus of theorem 1, while APPNP and the other explicit models receive the computational tool without the regime characterisation.

GAT[†] scope. Standard GAT uses A as a binary mask, so $\partial Z / \partial A_{ij} = 0$ on existing edges and S_c is undefined. Our GAT[†] variant modulates attention by the edge weight,

$$h'_i = \sum_{j \in \mathcal{N}(i)} \hat{A}_{ij} \cdot \alpha_{ij} \cdot W h_j, \quad (12)$$

restoring continuous differentiability with respect to $\hat{A}_{ij}$ and yielding tightness 0.99, AtkAdv $4.4\times$, $\tau = +0.36$. Architec-

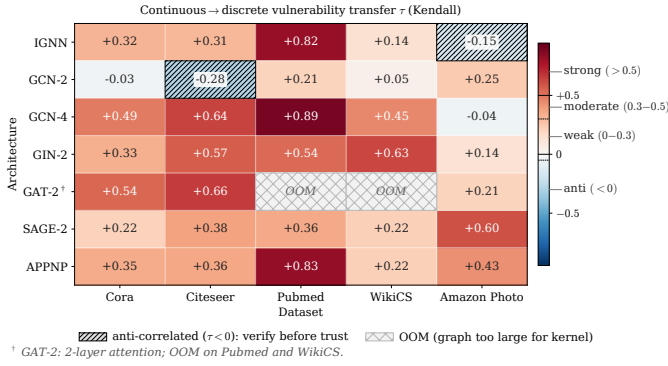


Fig. 5: Continuous-to-discrete transfer τ across 7 architectures and 5 datasets (see table VIII for precise values). Deeper models (GCN-4, APPNP) and attention (GAT[†]) consistently outperform shallow ones (GCN-2); gray: GPU OOM.

TABLE VIII: Continuous-to-discrete transfer τ (Kendall) across 5 datasets and 7 architectures (10 seeds each, 330 runs). Bold: best per dataset. —: GPU OOM.

Model	Cora	Citeseer	Pubmed	WikiCS	Am.Ph.
IGNN	+0.32±.04	+0.31±.04	+0.82±.02	+0.14±.04	-0.15±.03
GCN-2	-0.03±.03	-0.28±.07	+0.21±.01	+0.05±.03	+0.25±.03
GCN-4	+0.49±.02	+0.64±.02	+0.89±.01	+0.45±.10	-0.04±.06
GIN-2	+0.33±.04	+0.57±.03	+0.54±.33	+0.63±.08	+0.14±.07
GAT [†]	+0.54±.06	+0.66±.03	—	—	+0.21±.06
SAGE-2	+0.22±.11	+0.38±.18	+0.36±.10	+0.22±.04	+0.60±.04
APPNP	+0.35±.02	+0.36±.06	+0.83±.01	+0.22±.03	+0.43±.04

tures using binary adjacency masks (hard attention, max/min aggregation) and GATv2 [41]’s dynamic-attention masking fall outside the framework; the architecture-agnostic property applies only to continuous edge-weight-modulated message passing.

Cross-dataset transfer. Figure 5 reports continuous-to-discrete τ over 5 datasets and 7 architectures (330 runs). The two cold-cell regimes practitioners should distrust are shallow GCN-2 on sparse citation graphs, and IGNN on dense product graphs at 50-node subgraphs. The Amazon-IGNN case is diagnosable from the BFS cap $\kappa_{\text{sub}} \leq 0.20$ and the high average degree (~ 31) that flattens damage; explicit models on the same dataset still transfer cleanly. On heterophilic WebKB [42] GCN-4 attains $\tau = +0.21$ – $+0.44$, comparable to homophilic results.

Theorem 6’s sufficient condition holds for 47–62% of edge pairs (conservative, not tight); GCN-2 is the only architecture failing the bridge because 2-hop aggregates produce near-uniform first-order shifts. Gradient-based explainers [43] against the same discrete truth give $\tau = -0.07$ to -0.21 , since they optimise prediction fidelity not adversarial vulnerability; the theorem 7 bound is 1.4 – $5.9\times$ loose, tighter for shallower models. Per-edge v_{ij} correlates with endpoint degree only moderately, and reweighting S_c columns by degree does not improve rankings.

TABLE IX: AEGIS on IEEE power-flow benchmarks (10 seeds). Model quality: per-unit RMSE for $|V|$ and angle θ (std < 0.003). τ : Kendall correlation against brute-force N-1; P@10: top-10 precision. Case14–118 is the credible reported range; case300 is a scalability stress-test (model un-converged, not an operational point).

Case	N	$ E $	Model (p.u.)		AEGIS	
			$ V $	θ	τ	P@10
<i>Credible operating range</i>						
case14	14	20	.007	.020	$+.42_{\pm.19}$	$.74_{\pm.12}$
case30	30	41	.014	.024	$+.37_{\pm.17}$	$.68_{\pm.06}$
case57	57	78	.033	.059	$+.67_{\pm.09}$	$.66_{\pm.13}$
case118	118	179	.014	.076	$+.62_{\pm.11}$	$.81_{\pm.10}$
<i>Scalability stress-test (model un-converged)</i>						
case300 [‡]	300	411	.031	.394*	$+.72_{\pm.03}$	$.87_{\pm.06}$

[‡]200-node BFS subgraph; 1,000 training samples.

* θ RMSE $\approx 22.6^\circ$: not a valid operational data point.

VI. CASE STUDY: POWER FLOW CONTINGENCY

As a proof-of-concept cross-domain demonstration — not an operator-grade screening tool — we apply S_c to AC power flow. The conceptual correspondence is a structural isomorphism: the IFT-resolvent $(I - J_z)^{-1}$ at the GNN’s equilibrium is the linearised analogue of the post-contingency steady-state operator, so v_{ij} (continuous edge sensitivity) tracks N-1 contingency severity (discrete line removal) the way PTDF/LODF [38] linearise line trips for DC-PF screening. Recent GNN-based PF work [44], [45], [46], [3] optimises a fidelity objective; we take the vulnerability-analysis view.

Setup. ContractiveGCN-PF is an IGNN-class operator $F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ with spectral-norm-constrained W , trained on five IEEE test cases [36] with a 2-head readout $(\Delta|V|, \Delta\theta)$. Training data are 2,000 load samples per case (70–130% of nominal) from PandaPower’s [37] AC Newton-Raphson solver; inputs are binary adjacency and 5 bus features (slack/PV indicators, P , Q , $|V|$). For case14–118 we use full-graph dense analysis; for case300 a 200-node BFS subgraph. LODF is benchmarked off-spec on ℓ_2 voltage-angle truth (its native metric is line-flow change), so the ordering reflects the GNN’s training target.

table IX reports the envelope ratio and Kendall τ between S_c rankings and brute-force N-1; fig. 6 visualises case14. The envelope is tight in the PF domain (≈ 1.00), and on the credible range (case14–118) ranking correlation is $\tau = +0.37$ to $+0.62$ with P@10=0.66–0.81, identifying 7–8 of the top-10 critical lines. Case300 stress-tests the GNN’s learning capacity, not S_c scalability: at θ RMSE 0.394 p.u. ($\approx 22.6^\circ$) the surrogate has not learned AC physics; its apparent $\tau = +0.72$, P@10=0.87 are conditioned on an unconverged model and excluded from headline numbers. The matrix-free S_c pipeline itself scales to $N=7,650$ (fig. 4); improved PF surrogates would extend the demonstration into the operational range.

Baselines and stability. LODF (industry DC screening)

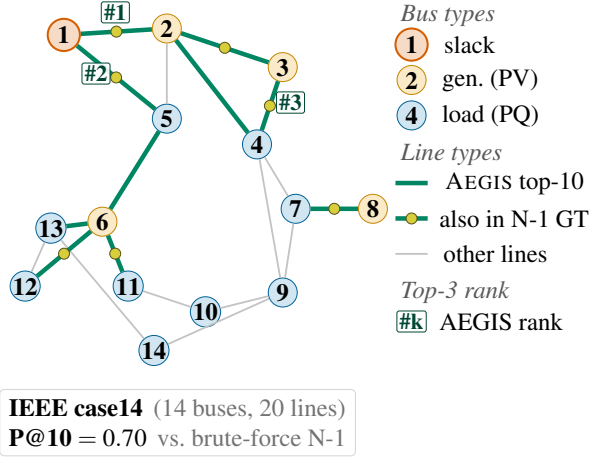


Fig. 6: AEGIS vulnerability ranking on the IEEE 14-bus benchmark (10-seed mean; $P@10 = 0.70$ for this run, consistent with the 0.74 ± 0.12 in table IX). Bus markers encode physical bus type (slack / generator / load); green edges are the top-10 by v_{ij} with rank tags (#1–#3); yellow midpoint dots mark edges that also appear in the brute-force N-1 top-10 (7 of 10). Top-ranked edges concentrate on the high-flow inter-area tie lines (1–2, 1–5, 7–8), recovering the single-contingency severity pattern from one closed-form SVD pass.

attains $\tau=0.44\text{--}0.58$ on credible cases, behind AEGIS’s $0.62\text{--}0.67$ on case57/118 (Wilcoxon $p<0.01$); CIs overlap on case14/30. The runtime trade-off is unfavorable for operations at this scale (LODF <0.13 s; brute-force N-1 0.1–2 s; AEGIS 2–23 s with training), and operators have impedance data, so the demonstration’s value is conceptual: a label-free GNN proxy recovers the dominant N-1 severity pattern from one closed-form SVD pass over a learned representation. v_1 captures one direction, so it surfaces multi-edge vulnerabilities only partially (top-5 overlap 40–64% with N-2 critical sets edge-wise, 7–18% pair-wise). Pairwise rank stability is $+0.40\text{--}+0.78$, higher on larger grids. *Binary adjacency beats admittance-weighted* ($P@10=0.81$ vs. 0.27 on case118): the topological discontinuity of a trip dominates magnitude information, a substantive cross-domain finding extracted directly by the GNN. Operator-grade deployment requires the extensions noted in the conclusion.

VII. RELATED WORK

We position AEGIS against four threads: structural attacks, certified robustness, implicit-network analysis, and influence-function sensitivity. Across all four, prior methods supply at most one of the three outputs we unify under a single matrix-free pass.

Adversarial attacks on graph structure. Nettack [1], Mettack [4], the BCD-family GR-BCD / PR-BCD [5], and related variants [47] optimise surrogate gradients to flip predictions, producing per-edge information but no closed-form globally maximally sensitive direction. Sober-look studies [15], [16] show prior defence benchmarks are vulnerable to evaluation

pitfalls; our defence-informed-masking ablation (section V-E) uses paired-Wilcoxon comparisons against random masking for this reason.

Certified robustness and defense. Convex relaxations [11], smoothing [6], [7], [8], collective certificates [9], and IBP [10] guarantee per-node robustness but do not identify which edges drive vulnerability. Empirical defenses [12], [14], [13] harden models without producing a vulnerability map. AEGIS complements certification with structural differentiation (section V-B); r_v is a first-order threshold, not a sound certificate (theorem 5).

Implicit networks and equilibrium analysis. DEQs [17], IGNN [19], and well-posedness analyses [26] address input sensitivity $\|\partial z^*/\partial x\|$ rather than structural sensitivity $\|\partial Z/\partial A\|$. AEGIS targets the latter via the resolvent $(I - J_z)^{-1}J_A$ for implicit models (theorem 1) and the unrolled Jacobian S_K for explicit ones (theorem 7).

Sensitivity and influence-function lineage. The closed-form structural derivative descends from influence-function-style implicit differentiation [25], extended here to adjacency under a constrained projection. GNN explainers [43] target prediction fidelity rather than adversarial vulnerability; AEGIS draws on classical matrix perturbation theory [29], [24].

VIII. CONCLUSION

AEGIS unifies per-edge vulnerability rankings, the SVD-optimal attack direction, and per-node first-order sensitivity radii under one constrained sensitivity matrix S_c , computed matrix-free for any GNN with continuous edge-weight-modulated message passing and supplemented by regime guarantees (theorem 1) for contractive implicit models.

Limitations. (i) Radii r_v are first-order sensitivity thresholds, not probabilistic certificates, and can be violated at large ε (theorem 5). (ii) Standard GAT and architectures using binary adjacency masks fall outside the framework; the seventh architecture is an edge-weighted GAT[†] variant (section V-F). (iii) On large citation networks, 50-node subgraph rankings correlate weakly with full-graph rankings ($\tau=0.16$ on Cora) and the matrix-free path requires $\kappa<1$; graph-level assessment uses the matrix-free pipeline, and explicit non-contractive GNNs use the dense or finite-difference fallback. (iv) IGNN accuracy (77.5% on Cora) lags explicit GNNs ($\sim 82\%$): the cost of the spectral-norm constraint required by (A2), so the formal-track guarantees apply to a model with degraded accuracy while higher-accuracy explicit models receive only the computational tool. (v) Power-flow $\tau=+0.37$ to $+0.62$ is a proof-of-concept; operator-grade deployment requires admittance-weighted edges, broader operating coverage, and thermal/voltage-violation severity rather than the ℓ_2 target used here. The same vulnerability map informs defence design (section V-E); code release will follow a 90-day stakeholder notification and a tiered-access protocol restricting attack-generation to institutional affiliates.

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